

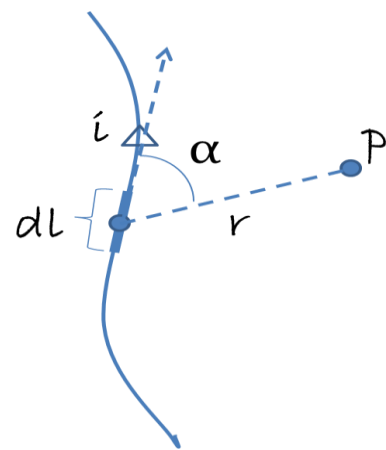
### The Biot-Savart Law:

An electric current flowing in a conductor, or a moving electric charge, produces a magnetic field, or a region in the space around the conductor in which magnetic forces may be detected. The value of the magnetic field at a point in the surrounding space may be considered the sum of all the contributions from each small element, or segment, of a current-carrying conductor. The Biot-Savart law states how the value of the magnetic field at a specific point in space from one short segment of current-carrying conductor depends on each factor that influences the field.

According to the Biot-Savart Law –

The magnitude of magnetic field ( $dB$ ) at a point produced around a conductor of small length due to the flow of current through it is –

- i) directly proportional to the amount of current. i.e.,  $dB \propto i$
- ii) directly proportional to the length of the conductor. i.e.,  $dB \propto dl$
- iii) directly proportional to the *sine* of the angle between the lines connecting the point from the mid-point of the conductor and the conductor itself. i.e.,  $dB \propto \sin\alpha$
- iv) inversely proportional to the distance between the point and the mid-point of the conductor. i.e.,  $dB \propto \frac{1}{r^2}$



So that, 
$$dB \propto \frac{idl \sin\alpha}{r^2}$$

$$\Leftrightarrow dB \propto \frac{idl \sin\alpha}{r^2}$$

$$\Leftrightarrow dB = K \frac{idl \sin\alpha}{r^2}$$

Here,  $K$  is a proportionality constant whose value depends on the nature of the medium and on the units of the quantities.

In S.I. unit for air medium or vacuum –

$$K = \frac{\mu_0}{4\pi}; \text{ Here, } \mu_0 \text{ is the magnetic permeability of vacuum. } \mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$$

Now, putting the value of  $K$ , then –

$$dB = \frac{\mu_0}{4\pi} \times \frac{idl \sin \alpha}{r^2}$$

For other medium –

$$dB = \frac{\mu}{4\pi} \times \frac{idl \sin \alpha}{r^2}$$

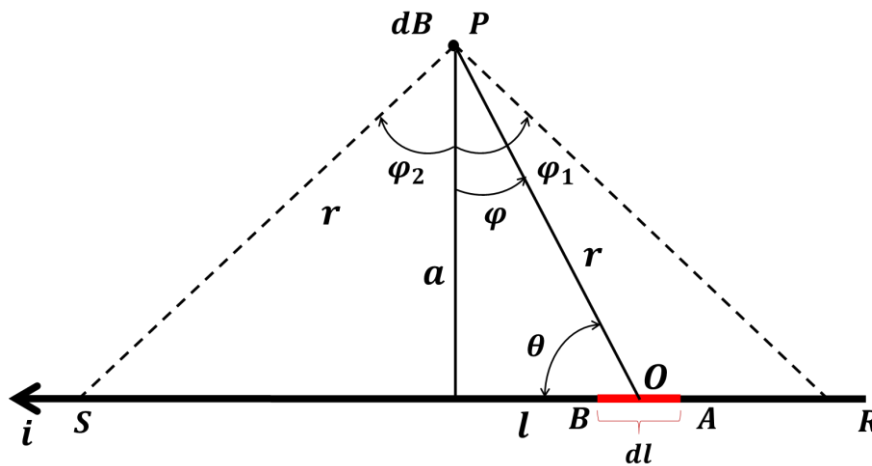
Vector form of the law –

$$\vec{dB} = \frac{\mu i}{4\pi} \frac{d\vec{l} \times \vec{r}}{r^3}$$

### Applications of Biot-Savart Law:

#### (a) Magnetic field due to a straight current carrying wire of infinite length:

Suppose  $RS$  is a long and straight wire and  $i$  ampere current is flowing through it. Magnetic induction or magnetic field at point  $P$  at a distance  $r$  from the wire is to be determined [Figure]



Let a very small section  $AB$  of that wire be taken. Let its length be  $dl$ . Suppose the distance from the center point  $O$  of that portion to be point  $P$  is  $r$  and angle  $\angle POB = \theta$ .

According to the Biot-Savart's law, the magnetic field at point  $P$  –

$$dB = \frac{\mu_0}{4\pi} \times \frac{idl \sin \theta}{r^2}$$

Now from figure, we get –

$$\sin \theta = \frac{a}{r} = \cos \phi$$

and

$$r = \frac{a}{\cos\varphi}$$

Also –

$$\tan\varphi = \frac{l}{a}$$

$$\Leftrightarrow l = a \tan\varphi$$

$$\therefore dl = a \sec^2\varphi d\varphi$$

Substituting  $r$ ,  $dl$  and  $\sin\theta$  in  $dB$  –

$$dB = \frac{\mu_0 i}{4\pi} \frac{a \cos\varphi \sec^2\varphi d\varphi}{\frac{a^2}{\cos^2\varphi}}$$

$$\therefore dB = \frac{\mu_0 i}{4\pi} \frac{\cos\varphi d\varphi}{a}$$

Magnetic field due to a whole conductor is obtained by integrating with limit from  $-\varphi_1$  to  $\varphi_2$ .  
 ( $\varphi_1$  is taken negative since it is Anticlockwise)

$$B = \frac{\mu_0 i}{4\pi a} \int_{-\varphi_1}^{\varphi_2} \cos\varphi d\varphi$$

$$\Leftrightarrow B = \frac{\mu_0 i}{4\pi a} |\sin\varphi|_{-\varphi_1}^{\varphi_2}$$

$$\Leftrightarrow B = \frac{\mu_0 i}{4\pi a} [\sin\varphi_2 - \sin(-\varphi_1)] = \frac{\mu_0 i}{4\pi a} [\sin\varphi_2 + \sin\varphi_1]$$

The wire is taken as infinite long, so that,  $\varphi_1 = \varphi_2 \cong 90^\circ$ ; then –

$$B = \frac{\mu_0 i}{4\pi a} [\sin 90^\circ + \sin 90^\circ] = \frac{\mu_0 i}{4\pi a} [1 + 1]$$

$$\therefore B = \frac{\mu_0 i}{2\pi a}$$

Magnetic field intensity produced due to a current carrying wire does not depend on the surrounding medium.

**(b) Magnetic field due to a circular current loop:**

Figure shows a circular loop of radius  $R$  carrying a current  $i$ , we would like to calculate the magnetic field  $B$  at a point  $P$  on the axis of the loop a distance  $Z$  from the center of the loop.

Let us first consider a small element  $dl$  as shown in the figure. The magnitude of the magnetic field  $dB$  produced by this element at the point  $P$  is given by Biot-Savart's law to be –

$$dB = dB = \frac{\mu_0}{4\pi} \times \frac{idl \sin 90^\circ}{r^2}$$

Where, the angle between the element  $dl$  and  $r$  is  $90^\circ$ .

$dB$  may be resolved into two components, one,  $dB_{||}$  along the axis of the loop, and another is the perpendicular component  $dB_{\perp}$ . The perpendicular component  $dB_{\perp}$  for every element  $dl$  will be equal and opposite to each other and the sum of all  $dB_{\perp}$  for the complete loop is zero. Only the components along the axis add and the total magnetic field  $B$  at point  $P$  is simply the sum of all  $dB_{||}$ . i.e.,

$$B = \int dB = \int dB_{||}$$

$$\text{Now, } dB_{||} = dB \cos \alpha = \frac{\mu_0 i}{4\pi} \frac{dl \cos \alpha}{r^2}$$

Hence,

$$B = \int dB_{||} = \int \frac{\mu_0 i}{4\pi} \frac{dl \cos \alpha}{r^2}$$

From figure, the relationships are –

$$r = \sqrt{R^2 + Z^2}$$

$$\text{and } \cos \alpha = \frac{R}{r} = \frac{R}{\sqrt{R^2 + Z^2}}$$

Substituting these values and we get,

$$B = \int \frac{\mu_0 i dl}{4\pi(R^2 + Z^2)} \frac{R}{\sqrt{R^2 + Z^2}} = \frac{\mu_0 i R}{4\pi(R^2 + Z^2)^{3/2}} \int_0^{2\pi R} dl$$

$$\Rightarrow B = \frac{\mu_0 i R}{4\pi(R^2 + Z^2)^{3/2}} 2\pi R$$

$$B = \frac{\mu_0 i R^2}{2(R^2 + Z^2)^{3/2}}$$

An important special case of this result is the magnetic field at the center of the loop ( $z = 0$ ). Equation then reduces to –

$$B = \frac{\mu_0 i}{2R}$$

For  $N$  turns of the circular wire –

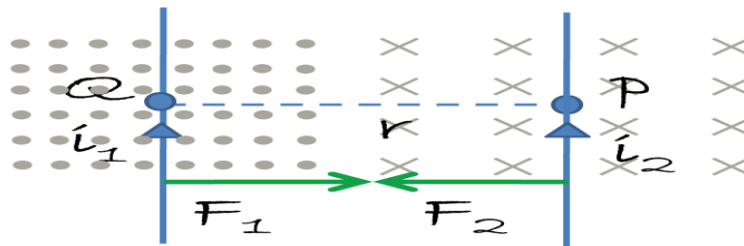
$$B = \frac{\mu_0 N i}{2R}$$

At distances from the loop,  $Z$  is much greater than  $R$ .  $R^2$  can, therefore, be neglected compared to  $Z^2$  in the denominator of the main equation then becomes –

$$B = \frac{\mu_0 i R^2}{2Z^3}$$

### Forces between two parallel current carrying conductors:

Figure shows two long parallel wires  $Q$  &  $P$  separated by distance  $r$ . Wire  $Q$  carrying current  $i_1$  and produces a magnetic field and the second wire  $P$  carrying current  $i_2$  and also produces a magnetic field. According to the Fleming's right hand rule, the wire  $P$  will be experienced the force toward the wire  $Q$  and vice-versa the  $Q$  wire will be experienced a force toward the wire  $P$ .



The magnitude of the forces –  $F_2 = \frac{\mu_0 i_1 i_2 l_2}{2\pi r}$  and  $F_1 = \frac{\mu_0 i_1 i_2 l_1}{2\pi r}$

The forces per unit length will be equal to each other.

$$\frac{F_1}{l_1} = \frac{F_2}{l_2} = F' = \frac{\mu_0 i_1 i_2}{2\pi r}$$

The important conclusion of these parallel current carrying wire is –

- For parallel current direction of both wire, attract each other
- For anti-parallel current direction of both wire, repel each other